

Mohamed Elsayed

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Research Objective.....

Developing general-purpose mechanisms for continual deep reinforcement learning agents, particularly resource-constrained embodied agents that learn in deployment

Education

Doctoral of Philosophy

University of Alberta, Canada

Computing Science. Advisor: A. Rupam Mahmood

GPA: 4.0/4.0

2022 - Present

Master of Science

University of Alberta, Canada

Computing Science. Advisor: A. Rupam Mahmood

GPA: 3.9/4.0

2019-2022

Bachelor of Science

University of science and technology, Zewail City, Egypt

Major: Communication and Information Engineering. Minor: Physics of the Earth and Universe.

GPA: 3.7/4.0

2014 – 2019

Publications

Conferences and Journals:

1. Vasan, G., **Elsayed, M.**, Azimi, S. A., He, J., Shahriar, F., Bellinger, C., White, M., & Mahmood, A. R. (2024). Deep policy gradient methods without batch updates, target networks, or replay buffers. *Neural Information Processing Systems (NeurIPS)*.
2. **Elsayed, M.**, Lan, Q., Lyle, C., & Mahmood, A. R. (2024). Weight clipping for deep continual and reinforcement learning. *Reinforcement Learning Journal*, 5(1), 2198–2217 (RLC).
3. **Elsayed, M.**, Farrahi, H., Dangel F., & Mahmood, A. R. (2024). Revisiting scalable Hessian diagonal approximations for applications in reinforcement learning. *International Conference on Machine Learning (ICML)*.
4. **Elsayed, M.**, & Mahmood, A. R. (2024). Addressing loss of plasticity and catastrophic forgetting in continual learning. *International Conference on Learning Representations (ICLR)*.
5. Zhou, M., Luo, J., Villella, J., Yang, Y., Rusu, D., Miao, J., ..., **Elsayed M.**, ..., & Wang, J. (2021). SMARTS: Scalable multi-agent reinforcement learning training school for autonomous driving. *Conference on robot learning (CoRL)*.
6. Elgammal, M. A., Elkhoully, O. A., Elhosary, H., **Elsayed, M.**, Mohieldin, A. N., Salama, K. N., & Mostafa, H. (2018). Linear and nonlinear feature extraction for neural seizure detection. *International Midwest Symposium on Circuits and Systems (MWSCAS)*.

Pre-prints and Workshops:

7. **Elsayed, M.**, Vasan, G., & Mahmood, A. R. (2024). Streaming deep reinforcement learning finally works. *arXiv preprint arXiv:2410.14606*.
8. **Elsayed, M.**, Vasan, G., & Mahmood, A. R. (2024). Deep reinforcement learning without experience replay, target networks, or batch updates. *NeurIPS Workshop on Fine-Tuning in Modern Machine Learning*.
9. **Elsayed, M.**, Mahmood, A. R. (2022) Utility-based perturbed gradient descent: An optimizer for continual learning. *NeurIPS Workshop on Optimization for Machine Learning*.
10. **Elsayed, M.**, Mahmood, A. R. (2022) HesScale: Scalable computation of Hessian diagonals. *NeurIPS Workshop on Higher-Order Optimization in Machine Learning*.
11. **Elsayed, M.**, Hassanzadeh, K., Nguyen, N. M., Alban, M., Zhu, X., Graves, D., & Luo, J. (2020). ULTRA: A reinforcement learning generalization benchmark for autonomous driving. *NeurIPS Workshop on Machine Learning for Autonomous Driving*.

Invited Talks

Streaming Deep Reinforcement Learning is Now Possible <i>Google DeepMind Vision Reading Group</i>	Jan 2025 <i>Online</i>
The Potential of Streaming Learning Agents <i>Scalable Next Generation AI for the Real World Retreat</i>	Dec 2024 <i>Sentosa, Singapore</i>
Streaming Deep Reinforcement Learning is Now Possible <i>RL Sofa at Mila</i>	Dec 2024 <i>Online</i>
Streaming Deep Reinforcement Learning is Now Possible <i>RLAI Meetings at University of Alberta</i>	Nov 2024 <i>Edmonton, Canada</i>
Streaming Deep Reinforcement Learning is Now Possible <i>Openmind Research Institute Retreat</i>	Oct 2024 <i>Banff, Canada</i>
Utility-based Representation Search <i>Tea Time Talk, University of Alberta</i>	Aug 2023 <i>Edmonton, Canada</i>
Towards Continual Learning Optimizers <i>RL Sofa at Mila</i>	April 2023 <i>Online</i>
Utility-based Perturbed Gradient Descent <i>Lifelong Reinforcement Learning Workshop</i>	February 2023 <i>Holetown, Barbados</i>

Work Experience

Huawei <i>Research Intern</i> - Worked on autonomous intersection navigation for the unprotected left turn problem with RL. - Co-authored a workshop paper in NeurIPS ML4AD.	Edmonton, Canada <i>May 2020 - Dec 2020</i>
Nara Institute of Science and Technology <i>Robotics Research Intern</i> - Worked on Semantic SLAM problem using deep learning techniques. - Worked on map in-painting techniques.	Nara, Japan <i>Aug 2018 - Sep 2018</i>
Valeo <i>Deep Learning Research Intern</i> - Worked on automotive-related projects (classification, semantic segmentation, and detection). - Handled new state-of-the-art network architectures and bench-marked on related works.	Smart Village, Egypt <i>Jul 2018 - Aug 2018</i>

Teaching

University of Alberta <i>Graduate Teaching Assistant</i> - CMPUT 174: Introduction to the Foundations of Computation - CMPUT 365: Introduction to Reinforcement Learning - CMPUT 340: Introduction to Numerical Methods	Edmonton, Canada <i>Sep 2019 - April 2020</i> <i>Sep 2022 - Dec 2022</i> <i>Jan 2024 - April 2024</i>
Zewail City of Science and Technology <i>Junior Teaching Assistant</i> CIE 417: Machine Learning	Cairo, Egypt <i>Sep 2018 - Dec 2018</i>

Service

- **Reviewer:** ICLR 2023, CoLLAs 2024, RLC 2024, NeurIPS 2024 (Top Reviewer), ICLR 2025, AISTATS 2025.
- **Panelist:** Creating (Continual) Agents Panel Discussion at Scalable Next Generation AI for the Real World Retreat in Sentosa, Singapore.
- **Evaluator:** European Lab for Learning & Intelligent Systems (ELLIS) 2024.

Skills

- **Programming Languages:** Python, C/C++
- **Software Libraries:** PyTorch, Jax
- **Technologies:** Git, Bash, $\text{\LaTeX}$